

NumPy Master Cheat Sheet (Placement & Interview Edition)

Designed as a last-minute revision guide for coding interviews, placements, ML, AI and Data Science.

1. Import

```
import numpy as np
```

2. Array Creation

Function	Purpose
<code>np.array()</code>	Create array
<code>zeros()</code>	All zeros
<code>ones()</code>	All ones
<code>empty()</code>	Uninitialized values
<code>full()</code>	Fill constant
<code>eye()</code>	Identity-like matrix
<code>identity()</code>	Identity matrix
<code>arange()</code>	Range with step
<code>linspace()</code>	Evenly spaced values

3. Array Properties

Property	Meaning
<code>shape</code>	Rows, columns
<code>ndim</code>	Dimensions
<code>size</code>	Total elements
<code>dtype</code>	Datatype
<code>astype()</code>	Datatype conversion

4. Indexing & Slicing

Positive: `arr[0]`

Negative: `arr[-1]`

Slice: `arr[1:4]`

Reverse: `arr[::-1]`

Row: `arr[1,:]`

Column: `arr[:,1]`

Fancy: `arr[[0,2,4]]`

Boolean: `arr[arr>50]`

Conditional: `np.where(cond,x,y)`

5. View vs Copy

view()	copy()
Shares memory	New memory
Fast	Independent
Less memory	More memory

6. Broadcasting Rules

Rule 1: Compare shapes from RIGHT to LEFT.

Rule 2: Dimensions must be equal OR one dimension must be 1.

Rule 3: Otherwise ValueError.

Memory Trick: Broadcasting stretches arrays; it does NOT copy them.

7. Universal Functions (ufuncs)

Arithmetic	Other
add subtract multiply divide	sqrt square power mod
abs exp log	sin cos tan

8. Aggregation

Function	Purpose
sum	Total
mean	Average
median	Middle
min/max	Smallest/Largest
argmin/argmax	Index of min/max
std	Standard deviation
var	Variance

axis=0 -> Columns

axis=1 -> Rows

9. Reshaping

Function	Purpose
reshape()	Change shape
reshape(-1)	Auto-calculate
flatten()	Copy to 1D
ravel()	View to 1D
transpose()/T	Swap rows & columns

resize()	Modify original
expand_dims()	Add axis
squeeze()	Remove axis size 1

10. Join & Split

Join	Split
concatenate()	split()
vstack()	vsplit()
hstack()	hsplit()
stack()	Adds new axis

11. Random Module

Function	Use
rand()	Uniform decimals
randn()	Normal distribution
randint()	Random integers
choice()	Random selection
shuffle()	Shuffle array
seed()	Reproducible randomness

12. Interview Differences

Topic	Difference
shape vs size	Dimensions vs total elements
ndim vs shape	No. of dimensions vs actual dimensions
view vs copy	Shared vs independent memory
flatten vs ravel	Copy vs view
max vs argmax	Value vs index
concatenate vs stack	Join vs new axis
vstack vs hstack	Vertical vs horizontal
Boolean indexing vs where	Filter vs replace
rand vs randn	Uniform vs normal

13. Complexity & Tips

Vectorization avoids Python loops.
 Prefer NumPy operations over for-loops.
 Broadcasting is memory efficient.

Use axis carefully.
Always verify shape before ML model input.

14. Top Placement Patterns

- Find max/min/mean of matrix
- Filter values using Boolean indexing
- Reverse arrays
- Reshape datasets
- Apply broadcasting
- Stack/split datasets
- Random train/test data generation

15. One-Line Memory Tricks

VSTACK = Vertical
HSTACK = Horizontal
ARG = Index
Flatten = Copy
Ravel = View
Axis 0 = Down Columns
Axis 1 = Across Rows
reshape(-1) = NumPy calculates
Broadcast = Stretch, don't copy
Boolean Indexing = Filter
np.where() = IF-ELSE for arrays